ESSnet Big Data

Specific Grant Agreement No 2 (SGA-2)

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Deliverable 4.7

Report about possible new statistical output based on (European) AIS data

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Prepared by:

Anke Consten (CBS, Netherlands)
Marco Puts (CBS, Netherlands)
Tessa de Wit (CBS, Netherlands)
Eleni Bisioti (ELSTAT, Greece)
Christina Pierrakou (ELSTAT, Greece)
Anna Bilska (GUS, Poland)
Michal Bis (GUS, Poland)
Øyvind Langsrud (SSB, Norway)

ESSnet co-ordinator:

Peter Struijs (CBS, Netherlands)
p.struijs@cbs.nl
Telephone : +31 45 570 7441
Mobile phone : +31 6 5248 7775

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1. Introduction

The aim of Work Package (WP 4) is to investigate whether real-time measurement data of ship positions, so-called Automatic Identification System (AIS) can be used 1) to improve the quality and internal comparability of existing statistics and 2) to produce new statistical products relevant to the European Statistical System (ESS). AIS is a mandatory collision avoidance system on ships. Each ship transmits a signal providing details of the ship's speed and location at frequent intervals. AIS data are picked up at numerous ground stations at ports and on the coast by various agencies globally. They are also picked up by satellites.

Five National Statistical Institutes participate in WP4: the national statistical institutes of The Netherlands (Work package leader), Denmark[1], Greece, Norway and Poland. WP4 is subdivided into two phases. Specific Grant Agreement 1 (SGA-1) focused on creating a common database, linking AIS-data to maritime statistics and constructing sea traffic analyses (February 2016-July 2017, See Annex 1 for a more detailed description). The current SGA-2 focuses on the calculation of emissions and future perspectives for AIS data as source data for new statistical output (August 2017-May 2018, see Annex 2 for a more detailed description). At the start of SGA-2 we had to change the scope of SGA-2, because good models for calculating emissions already existed and we could not get access to AIS data from the European Maritime Safety Agency (EMSA). Annex 3 describes the scope of SGA-2 after these changes. The current SGA will result in three new deliverables, of which this deliverable is the third one:

The current SGA will result in three new deliverables, of which this deliverable is the second one:

- Deliverable 4.6: "Deriving emissions using AIS" (to be delivered March 31st 2018)
- Deliverable 4.7: "Possibilities for new statistical output based on European AIS data" (to be delivered March 31st 2018)
- Deliverable 4.8: Consolidated report on project results (to be delivered May 31st 2018)

Besides these new deliverables, we also extend Deliverable 4.3 during SGA-2 with an extra chapter on the "Quality of satellite data from LuxSpace" (delivered January 31st 2018)

In deliverable 4.3 of SGA-1 we described the results of four Proof of Concepts (PoC's) on possibilities to improve *current* statistics by using AIS. In this deliverable 4.7 we will mostly focus on possibilities to construct *new* statistical products based on AIS. They partly overlap with the PoC's described in Deliverable 4.3. Furthermore, we wanted to involve other statisticians working on maritime statistics on thinking about the use of AIS for new statistical products. To this end, we sent out an expert questionnaire (in September 2017) on the use of AIS to maritime statisticians and all member countries of the ESSnet.

To begin with, possibilities for statistics on intra port distances are described in chapter 2. Then, possibilities on fluvio-maritime statistics are described in chapter 3. Chapter 4 compares the quality of Eurostat's average port-to-port distance matrix compared to Marine Traffic's model for estimating routes of ships and calculating the distance of a single journey. Chapter 5 describes the possibilities for

[1] Only in SGA-1, they withdraw from SGA-2

AIS as an economic indicator. Chapter 6 focuses on how AIS could be used to gain insight in effects of maritime disruptions or (changes in) regulations. Chapter 7 focuses on how visualization are indispensable for the understanding of data. Although visualizations are not a statistic as such, it usually does ask for a new type of skill, requiring an investment for most statistical offices. As such, it is important to be aware of the opportunities opened up by this investment. Chapter 7 provides an example of a visualization: sea routes taken by container giants performing transshipment. Finally, in chapter 8 the overall conclusions of this deliverable are provided. The results of the expert questionnaire are also described in chapter 8.

2. Possibilities for intra port distances

Port authorities do not always have a complete insight in the activities within their port. For example, they do not always know whether ships visit one or multiple terminals during one visit, rendering a higher intra-port journey distance. Using AIS, these journeys in the port, and thus the intra-port distances can be derived. Intra-port travel distances is a new statistical product that can be generated by processing AIS data. In the Proof of Concept (PoC), we developed an algorithm to calculate these intra-port journeys. The results from the expert questionnaire show that most of the NSI's are interested in a new product concerning intra-port travel distances by using AIS data.

The algorithm

Using the methodology developed for defining journeys (see Chapter 3, of deliverable 4.3), the point and time for a ship entering the port can be defined. Between the time of entering and departing the port, data points are selected. Then the data is median filtered over a 10-minute interval. This 10 minute interval renders sufficient valid measuring points in case a ship is at anchor and only transmits a signal every 3 minutes (see also deliverable 4.2, chapter 4). For each point and each successive point, the latitude and longitude are fed into the haversine function. The haversine function determines the great-circle distance between two points *on a sphere* given their longitudes and latitude:

$$\operatorname{hav}\left(\frac{d}{r}\right) = \operatorname{hav}(\varphi_2 - \varphi_1) + \cos(\varphi_1)\cos(\varphi_2)\operatorname{hav}(\lambda_2 - \lambda_1)$$

- Where hav is the haversine function:

$$\operatorname{hav}(\theta) = \sin^2\left(\frac{\theta}{2}\right) = \frac{1-\cos(\theta)}{2}$$

- d is the distance between the two points along a great circle of the sphere
- r is the radius of the sphere,
- φ_1, φ_2: latitude of point 1 and latitude of point 2, in radians
- λ_1, λ_2: longitude of point 1 and longitude of point 2, in radians

The algorithm was developed in Scala and can be found here:

https://github.com/mputs/WP4/blob/master/Portvisit2/src/main/scala/aisframe.scala

Further recommendations

In the future, it would be interesting to extend this algorithm in a way it can detect intra-port movements, where a ship that moves from one terminal to the other within the same port can be automatically detected. Another interesting aspect that could be investigated is the detection of anomalies in the movements of ships that could signal problems in the ports. Dutch Port authorities have also expressed interest in these topics. In general, they are interested in more information on inland waterway vessels visiting their ports.

3. Possibilities for fluvio-maritime transport

Inland waterway transport plays an important role for the transport of goods in Europe. More than 37,000 kilometres of waterways connect hundreds of cities and industrial regions. Twenty of the European member states have inland waterways, 13 of which are connected to each other. In 2016, 554 million tonnes of goods were transported on European inland waterways Germany and the Netherlands account for the largest transported volumes (54 and 49 billion TKM, respectively). The largest volume of traffic in Europe flows from the North Sea maritime ports (Rotterdam, Antwerp, Amsterdam, Ghent) to Germany and Switzerland, largely generated along the Rhine axis. Other important routes are the north-south axis from the Netherlands to Northern France via Belgium and the Danube route, flowing from west to east through Romania, Bulgaria, Austria, Slovakia, Hungary, and Croatia. Smaller inland networks are also exploited in the UK, Poland, Italy and the Czech Republic. [2]

In so-called fluvio-maritime transport, transport is partly performed on inland waterways (IWW) and partly on sea. For Eurostat, this has created discussion about whether fluvio-maritime transport should be reported as maritime transport, IWW transport, or both. Countries have non-harmonised approaches to report fluvio-maritime data in the maritime and IWW statistics. For example, Germany does not report transit transport through the Kiel Canal. Two countries report fluvio-maritime in both IWW and maritime statistics, while two other countries report fluvio-maritime only in maritime statistics. Some countries use the type of vessel while one country uses the port of loading/unloading to determine between IWW and maritime statistics [1].

Eurostat proposed that fluvio-maritime transport should be reported on the basis of the nature of the vessel (Working Group on Maritime Transport Statistics 23-24 May 2016). In other words, fluvio-maritime transport performed by an IWW vessel should be reported in the IWW statistics and fluvio-maritime transport performed by a seagoing vessel should be reported in the maritime transport statistics. The only exception to this, transport using seagoing vessels undertaken wholly on inland waterways, should be considered as IWW transport. If type of vessel information is unavailable in the source data, related information (such as port of loading/unloading) could be used to determine whether the fluvio-maritime transport is likely to be carried out by an IWW or a seagoing vessel.

Deliverable 4 describes the results of the PoC which investigates the extent of missing data on fluvio-maritime transport by maritime ships for the Netherlands. Fluvio-maritime transport by IWW ships is already part of the Dutch IWW statistics. However, the Netherlands does not report fluvio-maritime transport if it is performed by maritime ships. The definition used to define maritime ships are ships that have an IMO-number (IWW ships are ships with an EU-number). The extent of missing data on fluvio-maritime transport was not clear. We therefore investigated the number of maritime ships

travelling to Dutch IWW ports, as these are not part of both maritime and IWW statistics. We also investigated the number of maritime ships travelling over Dutch IWW to Belgian and German ports.

We investigated the number of maritime ships at four locations for a period of 9 days, (March 1-9th):

1. Port of Nijmegen (River Waal)
2. Port of Born/Stein (River Maas)
3. Location on the route to Germany: Hardinxveld-Giessendam (Boven-Merwede)
4. Location on the route to Belgium: Sas van Gent (Ghent-Terneuzen Canal)

Figure 1: used locations for analysis

Results showed that there were no maritime ships in the ports of Nijmegen and Born/Stein. We concluded from that there was no fluvio-maritime transport by maritime ships to Dutch IWW ports. However, analysing maritime ships route to Germany and Belgium by IWW, we found 33 ships going to Germany and 103 ships going to Belgium. If these ships also visited Dutch maritime ports first, they would be part of our maritime statistics. We found that of the maritime ships going to Germany and

Belgium, 78 ships loaded and/or unloaded goods in Dutch maritime ports around that period (February and March). The other 58 ships did not visit Dutch seaports in this time period, so the Netherlands has no further data on these ships. They probably came from the United Kingdom or other countries not accessible solely by IWW, constituting fluvio-maritime travel.

We concluded from this that fluvio-maritime transport (with loading/unloading in the Netherlands) probably constitutes a minimal part of transport statistics in the Netherlands. In itself this in an interesting finding, however we did not look at maritime ships travelling on inland waterways closer to maritime ports. Also, we did not investigate inland waterway ships travelling outside inland waterways (i.e. the North Sea). These cases also ask for further investigation. What is more, investigating fluvio-maritime travel is also interesting for other countries. For the group members of WP4, this idea is only relevant for Netherlands. However, France, Germany, Belgium and the United Kingdom could use further investigation to improve their current statistics on fluvio-maritime transport.

Further recommendations

From the perspective of traffic intensity, emissions and transit trade, it is interesting to further investigate these fluvio-maritime journeys. Aspects mentioned like maritime ships on inland waters closer to maritime ports, inland waterways ships travelling outside inland waterways, and focusing on fluvio-maritime transport in other countries demand further research. What we have not yet investigated is the relationship between maritime and IWW transport per se. As AIS can also be used to gain insight in this relationship because direct links might be seen. That is, if goods are loaded and unloaded for these ships around the same terminal and time, patterns in which goods are shipped from maritime ships to inland waterway ships and the other way around can be identified.

4. Port-to-port distance matrix based on model estimating shipping routes

The largest sea ports in Europe are located on the North Sea, with Rotterdam being the largest port in the EU. The concentration of ships in these few North Sea ports is higher than, for example, the Mediterranean ports. Five main markets can be distinguished in the European seaport system: the container market, the ro-ro market, the market for conventional general cargo, the liquid bulk market and the dry bulk market. Many factors and actors influence the way different types of cargo is routed through European ports to the hinterland. However, there are two main factors here: the connectivity of the port to the hinterland and the level of performance of the port itself [2]. Increasing traffic intensity and infrastructure requirements results in changing roles of ports.

The dynamics caused by all these factors requires a flexible way of calculating different aspects for each journey a ship undertakes. The emergence of AIS provides the opportunity to calculate all these aspects. When the interest lies in aggregating data for worldwide routes for a larger number of ships, or over a long time period, calculating these aspects for each journey separately might render inefficient processing. In these cases, it could be more efficient to model shipping routes. Modeling will also provide more insight in the factors determining the route a ship takes and insight in the patterns in shipping routes.

Marine Traffic has developed a tool for calculating shipping routes that is based completely on historical journeys (see here). See for references [3], [4], [5] and [6] Figure 2 shows an example of the journey from Piraeus (Greece) to Zeebrugge (Belgium).

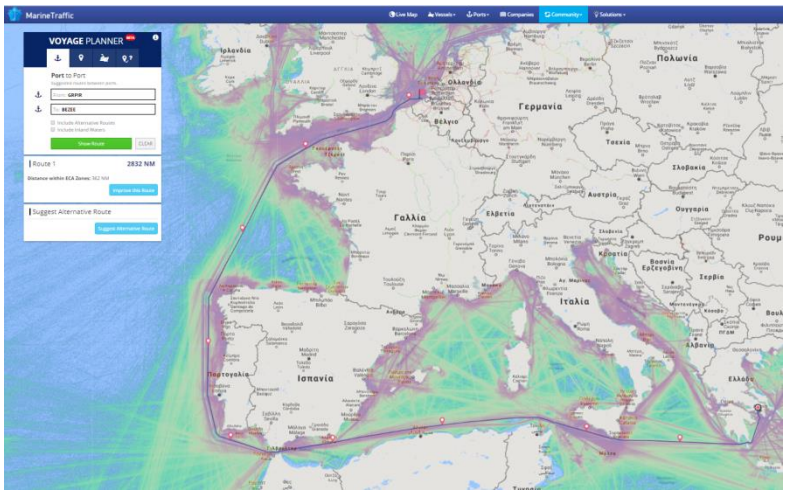

Figure 2: From Piraeus, Gr (GRPIR) to Zeebrugge, Be (BEZEE) distance 5308 km

From these routes the distance between two ports can be derived. The route from Piraeus to Zeebrugge for example is 5,245 km. The amount of extra information from past journeys make it interesting to compare the distance of some ship's journeys to distances that would currently be calculated by Eurostat by using Eurostat's port-to-port distance matrix. This matrix was developed in 2014 and also uses AIS for European journeys. However, the amount of historical data used is much lower and for extra-European travels AIS data is not used yet. Table 4.1 shows the distances in kilometers calculated for a specific journey by different methods: the distance determined over a number of journeys averaged by the Eurostat matrix (dEM), and by the Marine Traffic Tool (dMTt). The distance calculated for the actual journey based on the real time AIS data was also calculated by Marine Traffic (dMTa). The final two columns show the difference in distance between the methods by Eurostat Marine Traffic and the actual journey and the tool method from Marine Traffic compared to the actual journey.

from	to	shiptype	distance_Eurostat_ matrix (dEM)	distance_MT_ tool (dMTt)	distance_MT_ actual (dMTa)	difference dEM_dMTa	difference dMTt_dMTa
Barcelona (ES)	Piraeus (GR)	Container	2305	2165	2507	-8,1%	-13,7%
Tarragona (ES)	Rotterdam (NL)	Chem/oil	3687	3508	3565	3,4%	-1,6%
Aliaga (TR)	Amsterdam (NL	Chem/oil	6011	5710	5965	0,8%	-4,3%
Kokkola (FI)	Amsterdam (NL	GC/cont	2079	2443	2288	-9,1%	6,8%
St. Petersburg (RU	Rotterdam (NL)	Container	2155	2669	2080	3,6%	28,3%
Novorossiysk (RU)	Megara (GR)	Crude oil	1640	missing	1994	-17,7%	*
Zeebrugge (BE)	Piraeus (GR)	Carrier Vehicle	5284	5308	7039	-24,9%	-24,6%

Table 4.1 Distances per journey (km)

The difference between the methods can be quite large, up to a quarter of the distance. The differences are so large because some of the vessels selected are making intermediate stops (for example BEZEE, see figure 3) There is no structural over- or underestimating by either method.

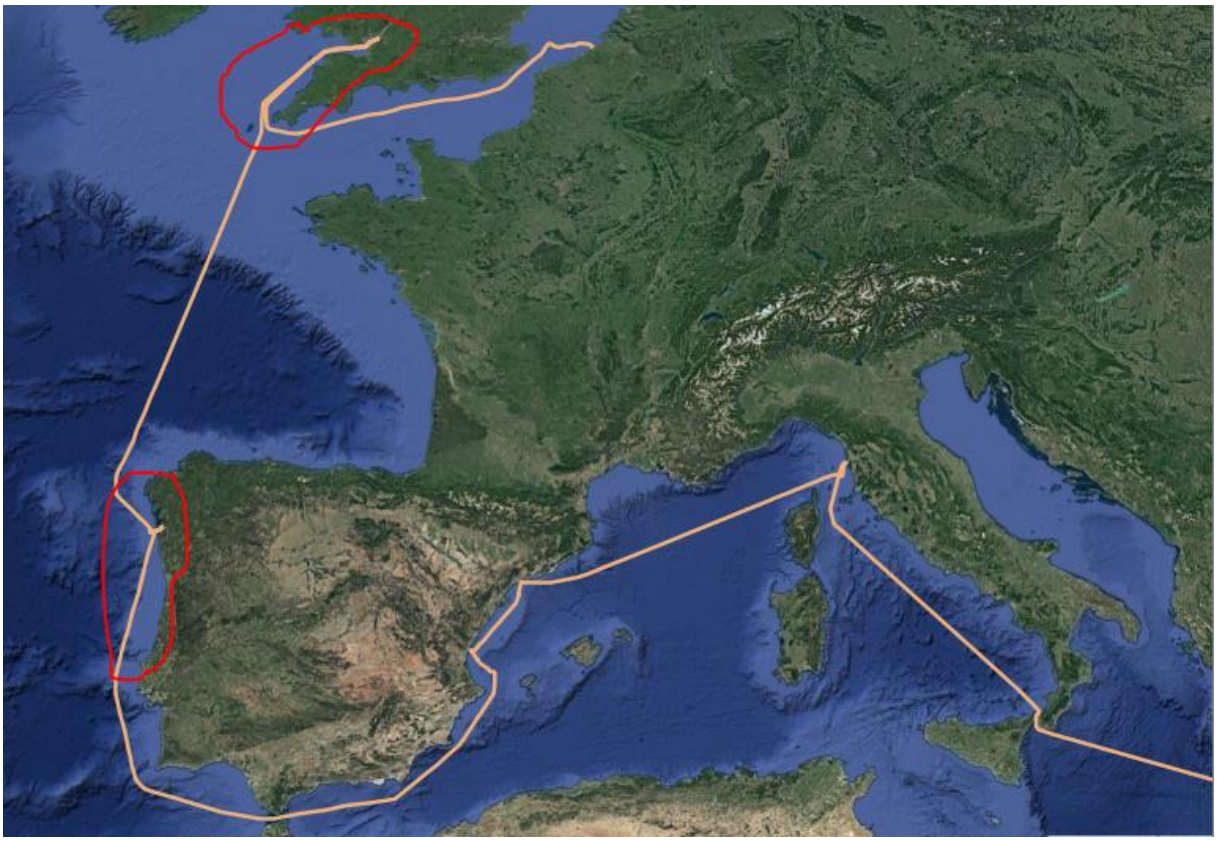

Figure 3: showing intermediate stops by one of the vessels

Here, only European travels were investigated. For worldwide travels we expect even higher deviations for Eurostat's method as Eurostat's method is not based on AIS there.

All in all, calculating the distance per single journey still seems most accurate (dMTa), as both averaging methods still differ quite a lot from the actual distance. Note however, that we only looked at a low number of samples. A larger number of samples is necessary justify conclusions. Yet, the method of calculating each journey separately requires an enormous amount of processing. We still think that fixed distance matrix that Eurostat uses is too limited. Mainly because it only uses a short period of historical AIS data, does not take into account important factors (e.g. type of ship, weather conditions) and does not use worldwide AIS data. Using a historically richer model like Marine Traffic's, and maybe adding factors like type of ship and weather conditions, could further improve the calculation of travelled distances of ships.

5. AIS as an economic indicator

The EU's international trade market highly depends on seaports, as 74% of goods imported from and exported to the rest of the world and 37% of intra-EU trade transit through seaports [7]. To get up-to-date information on the development of international trade, and with that economic development, an

analysis of activity in the seaports would be a valuable insight. Part of this information could be based on AIS data.

In other words, AIS could be used as a fast indicator of economic development. The expert questionnaire showed that most NSI's think it is possible to use AIS data as a new data source for now-casting of economic time series. As AIS can be available almost in real-time, it can help monitoring developments in transport and destinations of certain types of goods fast. Research has to be performed to identify which aspects cohere with economic developments (e.g. the increase in the number of a certain type of ship visiting a port). An idea was to generate time series of activity for ships in ports and then check for correlations with economic time series, like trade volume. We wanted to perform this experiment with national AIS data from Denmark, because Statistics Denmark has access to historical AIS data, enabling time series analysis. Unfortunately, because of the withdrawal from Denmark out of the ESSnet we could not investigate this further, but it will be an interesting investigation for the future. Going a step further would be to even now-cast some economic time series for which we need to have access to real-time AIS data in the future.

AIS could also be used as an indicator of economic development on an international level. By analyzing patterns in shipping, shifting roles in the maritime community can be identified. Here, specific ports or countries might gain or lose shares in the shipping world or develop a specialization in specific goods or phenomena, like transshipment.

6. Insight in effects of disruptions or regulations

Multiple parties have shown interest in gaining insight into the effects of disruptions or (changed) regulations by means of AIS. This was also shown by the expert questionnaire. For example a shift in traffic patterns caused by port closure could be found by using AIS. Another example are speed regulations in for example emissions zones.

In this work package the average speeds in Europe are calculated. See Deliverable 4.6. We could adjust this algorithm by calculating the speed of ships based on the distances and time travelled and compare the calculated speeds with the speeds available in the AIS data.

We could also develop a speed grid of Europe for the maximum speed travelled. From this we could determine if ships adhere their speed to these regulations. The other way around is also possible: to check if the ships adjust their speed in these regulation zones.

7. Visualizations

Visualizations are indispensable when it comes to gaining insight in data, especially in big data like AIS. AIS can be used for a wide range of visualizations that can bring understanding in a various different aspects of shipping. We will provide an example here of Dutch work combining different sources of data into a visualization to inform a broader public on transshipment by container giants.

Besides investigating AIS, Statistics Netherlands is currently investigating data from Dutch customs on transshipment. The data contains information on the containers that maritime ships dropped off on quays in Dutch ports and were picked up by other ships to transport to another port. This research focused on container giants specifically. These container giants have a cargo capacity of

more than 10 thousand TEU (the standard capacity unit for container ships). They can unload on average nearly 3 thousand loaded containers per visit onto Dutch quays. In 2016, over 900 of these giants called at the Port of Rotterdam to unload containers. The routes of the ships and thus all the ports that a ship visits can be shown using AIS data. Figure 2 shows how two container giants, one after the other, depart from China, pass the Suez Canal and reach the Port of Rotterdam. A substantial number of containers are unloaded at the quays of Rotterdam and then transshipped to other ports by other ships. The routes of these connecting ships is also shown. For certain longer routes there is no information available due to missing data. Therefore, these routes have been estimated and may deviate somewhat from the actual route taken. The dynamic visualization is available on the research site of Statistics Netherlands: http://research.cbs.nl/AIS_transshipment.

Here it can be seen that the first container giant makes several stops at other European ports, where some of the containers are unloaded. The information on where a ship call before or after a certain port is not always available in the maritime data. Because the route is clear from AIS, it can be derive from that.

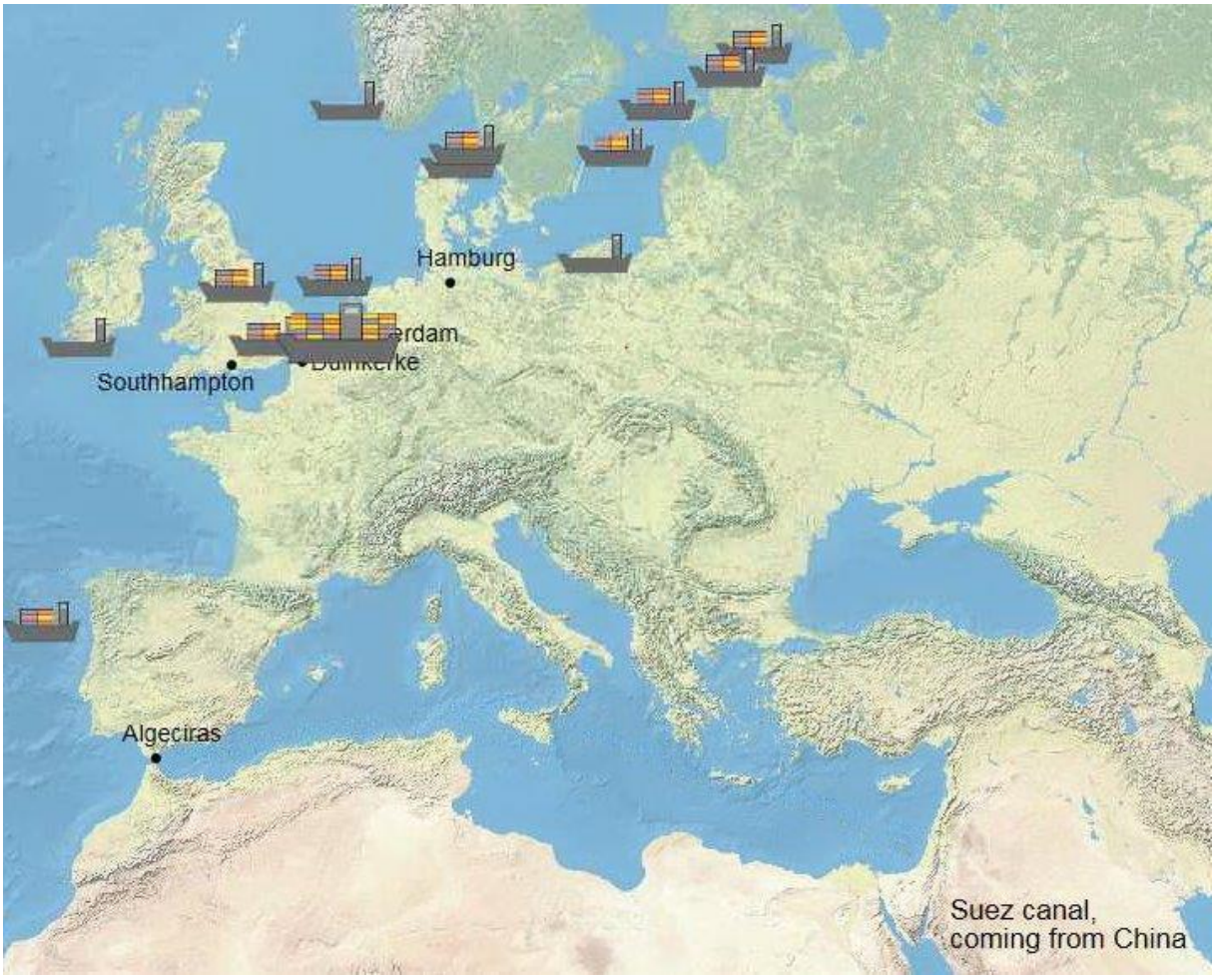

Figure 4: screenshot visualisation of container giants and connecting vessels and their sea routes

This visualization helped the general Dutch public gain understanding in the concept of transshipment and the pattern of work by container giants in the form of a sort of story-telling publication.

Further recommendations

Depending on the goal and public of a visualization, visualizations of ship routes can focus on a single ship when the goal is to be used to tell a story, like in the example above or it can focus on a bigger number of ships. For example on averages of a certain type or size of ships. The amount of geographical detail can also be varied. In the example shown, the focus is on the route from the Suez Canal (not from China due to technical difficulties) and ports. It is also interesting to zoom in on certain ports or another option is to investigate the possibilities for deriving missing information on goods, by getting the kind of goods depending on the quay the ships load or unload.

The visualization of AIS data can be used to monitor developments in transport and destinations of certain types of goods. It enables comparisons on an international level and helps identifying patterns. It will be clearly visible, for example, when more container ships are coming to Rotterdam. It potentially could be used as a fast economic (see chapter 5) or environmental indicator (e.g., CO_2 emissions).

8. Overall conclusion

The results of the expert questionnaire we administered among maritime statisticians and all member countries of the ESSnet in September 2017 showed that most of the NSI's expect that the use of AIS data could prove useful for maritime statistics. That is, they expect AIS to improve statistics on port visits, as well as calculation of routes and the port-to-port distance matrix. Importantly, maritime experts foresee the use of AIS data to open up opportunities to enhance timeliness and accessibility of maritime statistics. The questionnaire also shows that most NSI's believe in AIS as a source for new statistical products.

In the current deliverable we focused on these new products, compared to the previous deliverables in which we focused on starting to work with AIS and improving existing statistical products. This deliverable is more forward looking with many new ideas for new statistical products based on AIS. Some built on (preliminary) results of Proof of Concept (PoC) we performed. For example, in deliverable 4.3 it was shown that we can calculate intra-port travel distances by using AIS data. For future work, we would like to extent this algorithm for detecting intra-port movements, that is, an algorithm automatically signalling movement of ships between terminals and identification of these terminals within a port.

Another PoC showed that AIS data is also useful to gain insight into fluvio-maritime transport, since the extent of this transport mode is not always clear for every country. Focusing on fluvio-maritime journeys will also provide insight in the relationship between maritime and inland waterway transport Furthermore, it will also contribute to knowledge on transit trade, traffic intensity, and emissions.

Other findings show that using a more concise model, like the one of Marine Traffic, could improve the calculation of ships' travelled distances between ports. The current calculation used by Eurostat depends on a fixed distance matrix. However, this method does not take into account important factors known to influence the route of a ship, such as type of ship, weather and seasonal conditions.

This deliverable also describes some new ideas for future research. Some of them might turn out to be successful and some may not, but all are worth a try. The most promising idea is using AIS data as a fast economic indicator. Fast economic indicators are the holy grail in official statistics. Finding "fast" data sources that are closely related to economic development are crucial for creating these indicators. AIS could be one of the candidates as a proxy for international trade, signalling changes in maritime activity relating to international trade, and with that economic developments. Most NSI's agree that AIS data could provide a useful new data source for now-casting of economic time series.

Another aspect that was reported in the expert questionnaire to be interesting for future research is the detection of anomalies in the movements of ships. These anomalies could signal problems in the ports or (changed) regulations. Dutch Port authorities have also expressed interest in these topics, so In the expert questionnaire we also conclude that multiple parties are interested in gaining insight into the effects of disruptions or (changed) regulations by means of AIS. One relatively easy achievable product here is to produce the speed grid of Europe for the maximum speed travelled, signaling whether ships comply with these speed regulations.

Similar to the previous example, all of these ideas of using AIS data for improving/new official statistics can be combined with powerful visualizations. These can provide more insight in the developments in transport and destinations of certain types of goods. It also facilitates comparisons on an international level and helps identifying patterns.

The results described in this deliverable shows the potential wealth of AIS data to generate new statistical products. Although more investigation and (nearly) real time and world-wide AIS data is needed to result in better new statistical products. In the final deliverable (deliverable 4.8) for this WP, we will discuss which of these ideas could be implemented for the ESSnet as a whole, for example by using one standardised European AIS source and methodology.

9. References

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Annex 1: SGA-1 of WP4 in more detail

Work package number	4	Start date: End date:			1.2.2016 31.7.2017		
Title	**AIS Data**						
Partner/co-beneficiary	**NL** **120**	DK 60	EL 40	NO 67	PL 40		

Aim of this work package is to investigate whether real-time measurement data of ship positions (measured by the so-called AIS-system) can be used 1) to improve the quality and internal comparability of existing statistics and 2) for new statistical products relevant for the ESS. Improvement of quality and internal comparability can be obtained e.g. by developing a reference frame of ships and their travels in European waters and then linking this reference frame, by ship number, to register-based data about marine transport from port authorities. These linked data can then be used for emission calculations. New products can be developed for e.g. traffic analyses. The added value of running a pilot with AIS-data at European level is that the source data are generic word wide and data can be obtained at European level. Challenges ahead with this dataset are: obtaining the data at European level, processing and collecting the data in such way that they can be used for multiple purposes, and visualising the results. A part of this work package is also to look into AIS analyses done by others and to investigate the possibility of obtaining already processed data as input for creating comparable official statistics. Especially it is important to make contact with other public authorities. This work package may require data acquisition in collaboration with Eurostat.

Methodological, quality and technical results of the work package, including intermediate findings, will be used as inputs for the envisaged WP 8 of SGA-2, in case SGA-2 will be realised. When carrying out the tasks listed below, care will be taken that these results will be stored for later use, by using the facilities described at WP 9.

Task 1 – Data access.

AIS-data are available for national territories and the entire European territory. For example, AIS-data in the Netherlands are provided by Rijkswaterstaat (a government agency which is part of the Ministry of Infrastructure and the Environment). It is expected that similar agencies in other countries have the AIS-data for their national territories. At the European level, a dataset of AIS data is available at the European Maritime Safety Agency (EMSA). The advantage of using one AIS-dataset for the entire European territory is a) a better comparison of international traffic between the countries and b) more synergy as all participating countries work on the same dataset. A disadvantage is that these data are stored by private companies and handling fees have to be paid. Aim of this task is to decide how European data could be

used for this project, to investigate the possibilities of acquiring data from EMSA (to be coordinated with Eurostat) and, if European data are too costly or too hard to obtain, how national datasets can be obtained.

This task will involve:

- Exploration of the possibilities to collect the data at a European (or worldwide) level.

Participants: NL, DK, EL, NO, PL

Task 2 – Data handling

Aim of this task is to process and store the data in such a way that they can be used for consistent multiple outputs, like

- linking AIS-data with data from port authorities
- traffic analyses
- Inference of journeys from AIS data.

Key elements of this task are:

- which programming language and environment should be used for transformation
- where will the data be processed (in each NSI, by NSIs at European level, by data holders)
- how can we create an environment which is easily accessible for all partners

Participants: NL, NO, PL, and possibly DK (if national sources are used)

Task 3 – Methodology and Techniques

Develop traffic statistics: Linking with data from port authorities.

AIS-data may be linked to data from port authorities. Added value of linking AIS-data to data from port authorities is that the same reference population (= ship number) is used in all harbours. As the journeys and harbour visits of ships can be derived from AIS this linking provides the ESS information about the origin/destination of the cargo, too.

Aims of this task are:

- to build a reference frame of ships in European water (based on AIS-data)
- to find out how data from port authorities can be linked to AIS-data
- to check whether information improves the quality of current statistical outputs and provides more information about the origin/destination of the cargo.

Participants: NL, NO, PL

Traffic analyses

The number of ships during a certain time interval at certain coordinates (like inland waterways or at certain points at sea) can be calculated by AIS-data. This possibility will be explored because this information could be interesting for traffic analyses and economic analyses.

Aims of this task are:

- calculate the number of ships at certain coordinates
- visualise the results to analyse variations in time

Participants: NL, NO, PL, EL

Estimate emissions (envisaged under SGA-2)

This task involves following individual vessels through time. Consequently we can infer the journeys from the data. Combined with a model to estimate the emission of vessels (which depends on travel distance, speed, draught, weather conditions and characteristics of the vessel itself), emissions of e.g. CO_2 and NO_x can be estimated per ship and per national territory.

An advantage of doing these analyses on a European scale– instead of the national level – is that more precise estimates for emissions at national territories can be made.

Aim of this task is 1) to infer journeys from AIS-data, 2) visualise the results, 3) combine these journeys with a model to calculate emissions and 4) estimate the impact of carrying out these calculations at the European level on the quality of emissions calculations.

Task 4 – Future perspectives (envisaged under SGA-2)

Aim of this task is to summarise the project results and perform a qualitative cost-benefit analysis of using AIS-data for official statistics. These analyses should include aspects like sustainability of the data source, possibilities of improving international comparability, possibilities of data sharing (at micro- or aggregated level), quality improvement of current statistics and a sketch of a possible statistical process and needed infrastructure.

Deliverables (SGA-1 only):

4.1	Report on creating a database with AIS-data for official statistics: possibilities and pitfalls	Month 6
4.2	Report about deriving harbour visits and linking data from port authorities with AIS-data	Month 12
4.3	Report about sea traffic analyses using AIS-data	Month 18

Milestones (SGA-1 only):

4.4	Progress and technical report of first internal WP-meeting	Month 4
4.5	Progress and technical report of second internal WP-meeting	Month 9

Annex 2: original scope SGA-2 of WP4 in more detail

Work package number	4	Start date: End date:			1.8.2017 31.5.2018
Title	**AIS Data**				
Partner/co-beneficiary **(person days)**	**NL** **88**	DK 53	EL 66	NO 21	PL 35

Description of the work package

Aim of this work package is to investigate whether real-time measurement data of ship positions (measured by the so-called AIS-system) can be used 1) to improve the quality and internal comparability of existing statistics and 2) for new statistical products relevant for the ESS. Improvement of quality and internal comparability can be obtained e.g. by developing a reference frame of ships and their travels in European waters and then linking this reference frame, by ship number, to register-based data about marine transport from port authorities. These linked data can then be used for emission calculations. New products can be developed for e.g. traffic analyses. The added value of running a pilot with AIS-data at European level is that the source data are generic worldwide and data can be obtained at European level.

Challenges ahead with this dataset are: obtaining the data at European level, processing and collecting the data in such way that they can be used for multiple purposes, and visualising the results. A part of this work package is also to look into AIS analyses done by others and to investigate the possibility of obtaining already processed data as input for creating comparable official statistics. Especially it is important to make contact with other public authorities. This work package may require data acquisition in collaboration with Eurostat.

Methodological, quality and technical results of the work package, including intermediate findings, will be used as inputs for WP 8 of SGA-2. When carrying out the tasks listed below, care will be taken that these results will be stored for later use, by using the facilities described at WP 9.

SGA-2 of WP 4 will deliver the products below:

Task 3 – Methodology and Techniques

Estimate emissions

This task involves following individual vessels through time. Consequently we can infer the journeys from the data. Combined with a model to estimate the emission of vessels (which depends on travel distance, speed, draught, weather conditions and characteristics of the vessel itself), emissions of e.g. CO_2 and NO_x can be estimated per ship and per national territory.

Estimation of emissions based on only AIS data is impossible. At least, we need to know what the emission should be given, a.o., draught, speed and weather conditions. There are several sources that can be used to get information about the emission of vessels. For example Lloyds Register of Shipping.

Also, all ships have to register their emissions, so there is data available at the freight ship companies. A dataset for the same period as the AIS dataset could be used to model the emissions based on the variables mentioned above. We assume getting these needed data for free for this pilot.

Furthermore, satellite data is available for emissions (see, for instance http://www.globemission.eu), which shows very clearly the maritime routes. This satellite data could be used as a more direct source for measuring emissions and maybe for testing our model. Another possibility to test the model is to compare the estimated emissions by the model with the real fuel purchase or information coming from vessel's oil record book.

An advantage of doing these analyses on a European scale– instead of the national level – is that more precise estimates for emissions at national territories can be made.

Aim of this task is to develop and test a methodology for estimating vessel emissions based on AIS data by:

1) inferring journeys from AIS-data
2) visualising the results
3) investigating methodology for calculating emissions
4) combining these journeys with a model to calculate emissions
5) Testing the model and
6) estimating the impact of carrying out these calculations at the European level on the quality of emissions calculations.

Participants: Norway, Netherlands , Denmark, Greece, Poland

Task 4 – Access to and analysing AIS data from EMSA

AIS-data are available for national territories and the entire European territory. In this work package we use European AIS data from Dirkzwager. But for the future we would like to get free European AIS data. That is why we try to get AIS data at the European level from the European Maritime Safety Agency (EMSA).

In SGA-2 we focus on getting the European AIS data from EMSA and compare these two sources (AIS data from EMSA and Dirkzwager) on their coverage and quality. We will assess the quality of both sources by applying the quality framework. We also describe the strengths and weaknesses of both sources compared with existing maritime data.

This work package may require data acquisition in collaboration with Eurostat, because we would like Eurostat to apply for the data at EMSA's. Eurostat already received AIS data from EMSA for other projects and Eurostat can apply EMSA data for all the partner countries once. This will be better than applying for each country separated. If we do not get access to the EMSA data we cannot process this task and we

cannot deliver deliverable 4.7, but it has not any consequences for the other deliverables in this WP. Not getting AIS data from EMSA means that we cannot get AIS data on the European level for free.

Participants: Netherlands , Denmark, Greece, Poland

Task 5 – New statistical output

Aim of this task is to explore possibilities of new statistical products (for example interport statistics) by using AIS data. This task also includes analysing and elaborating scenarios for production of European and national statistics based on one single European data source of AIS data.

Participants: Norway, Netherlands, Denmark, Greece, Poland

Task 6 – Future perspectives

Aim of this task is to produce a consolidated report summarising the contents and the outcomes of WP 4. This report also includes a cost-benefit analysis of using AIS-data for official statistics. The report should also include aspects like sustainability of the data source, possibilities of improving international comparability, possibilities of data sharing (at micro- or aggregated level), possibilities of meeting the needs of both European and national statistics by one European AIS database, quality improvement of current statistics and a sketch of a possible statistical process and needed infrastructure**,** including technical skills required to generate statistical outputs from the source data.

Participants: Norway, Netherlands, Denmark, Greece, Poland

Deliverables (SGA-2 only):

4.6	Report on estimating emissions. This report will describe the investigated methodology for calculating emissions and the reason why we choose for a certain methodology. The report also describes the created model itself (inclusive other needed data sources) and the results of testing the model. Finally this report also describes the impact of carrying out these calculations at the European level on the quality of emissions calculations.	Month 13
4.7	Report about the results of comparing the quality and coverage of the European AIS data from Dirkzwager and EMSA (by applying the quality framework). Also the strengths and weaknesses of both sources will be compared with existing maritime data.	Month 15
4.8	Report about possible new statistical output based on European AIS data. The report also describes analysed and elaborated scenarios for production of European and national statistics based on one single European data source of AIS data.	Month 16
4.9	Consolidated report on project results including a cost-benefit analysis of using AIS-data for official statistics.	Month 17

Milestones (SGA-2 only):

4.10	Progress and technical report of first internal WP-meeting	Month 9
4.11	Progress and technical report of the second internal WP-meeting	Month 14

Annex 3: adjusted scope SGA-2 of WP4 in more detail

Work package number	4	Start date: End date:			1.8.2017 31.5.2018
Title	**AIS Data**				
Partner/co-beneficiary **(person days)**	**NL** **88**	DK 53	EL 66	NO 21	PL 35

Description of the work package

Aim of this work package is to investigate whether real-time measurement data of ship positions (measured by the so-called AIS-system) can be used 1) to improve the quality and internal comparability of existing statistics and 2) for new statistical products relevant for the ESS. Improvement of quality and internal comparability can be obtained e.g. by developing a reference frame of ships and their travels in European waters and then linking this reference frame, by ship number, to register-based data about marine transport from port authorities. These linked data can then be used for emission calculations. New products can be developed for e.g. traffic analyses. The added value of running a pilot with AIS-data at European level is that the source data are generic worldwide and data can be obtained at European level.

Challenges ahead with this dataset are: obtaining the data at European level, processing and collecting the data in such way that they can be used for multiple purposes, and visualising the results. A part of this work package is also to look into AIS analyses done by others and to investigate the possibility of obtaining already processed data as input for creating comparable official statistics. Especially it is important to make contact with other public authorities. This work package may require data acquisition in collaboration with Eurostat.

Methodological, quality and technical results of the work package, including intermediate findings, will be used as inputs for WP 8 of SGA-2. When carrying out the tasks listed below, care will be taken that these results will be stored for later use, by using the facilities described at WP 9.

SGA-2 of WP 4 will deliver the products below:

Task 3 – Methodology and Techniques

Estimate emissions

This task involves following individual vessels through time. Consequently we can infer the journeys from the data. Combined with a model to estimate the emission of vessels (which depends on travel distance, speed, draught, weather conditions and characteristics of the vessel itself), emissions of e.g. CO_2 and NO_x can be estimated per ship and per national territory.

Estimation of emissions based on only AIS data is impossible. At least, we need to know what the emission should be given, a.o., draught, speed and weather conditions. There are several sources that can be used to get information about the emission of vessels. For example Lloyds Register of Shipping.

Also, all ships have to register their emissions, so there is data available at the freight ship companies. A dataset for the same period as the AIS dataset could be used to model the emissions based on the variables mentioned above. We assume getting these needed data for free for this pilot.

Furthermore, satellite data is available for emissions (see, for instance http://www.globemission.eu), which shows very clearly the maritime routes. This satellite data could be used as a more direct source for measuring emissions and maybe for testing our model. Another possibility to test the model is to compare the estimated emissions by the model with the real fuel purchase or information coming from vessel's oil record book.

An advantage of doing these analyses on a European scale– instead of the national level – is that more precise estimates for emissions at national territories can be made.

Aim of this task is to investigate existing models for deriving emissions and describe how we could implement these models in our statistical processes. We also will investigate possibilities for improving emission statistics by using AIS data.

Participants: Norway, Netherlands , , Greece, Poland

Task 4 – Access to and analysing AIS data from EMSA

AIS-data are available for national territories and the entire European territory. In this work package we use European AIS data from Dirkzwager. But for the future we would like to get free European AIS data. That is why we (together with Eurostat) try to get AIS data at the European level from the European Maritime Safety Agency (EMSA).

In SGA-2 we focused on getting the European AIS data from EMSA. Unfortunately, we did not succeed in getting access to data from EMSA During SGA-2. We did obtain satellite data from LuxSpace covering the Mediterranean Sea for the same period as the Dirkzwager data. So we could assess the quality of satellite data by comparing it to the DZ data and the national AIS data from Greece.

Participants: Netherlands, Greece, Poland

Task 5 – New statistical output

Aim of this task is to explore possibilities of new statistical products (for example interport statistics) by using AIS data. This task also includes analysing and elaborating scenarios for production of European and national statistics based on one single European data source of AIS data.

Participants: Norway, Netherlands, Greece, Poland

Task 6 – Future perspectives

Aim of this task is to produce a consolidated report summarising the contents and the outcomes of WP 4. This report also includes a cost-benefit analysis of using AIS-data for official statistics. The report should also include aspects like sustainability of the data source, possibilities of improving international comparability, possibilities of data sharing (at micro- or aggregated level), possibilities of meeting the

needs of both European and national statistics by one European AIS database, quality improvement of current statistics and a sketch of a possible statistical process and needed infrastructure, including technical skills required to generate statistical outputs from the source data.

Participants: Norway, Netherlands, Greece, Poland

Deliverables (SGA-2 only):

4.6	Report on estimating emissions. This report will describe the results of investigating existing models for calculating emissions. The report also will describe the results of investigating possibilities for improving emission statistics by using AIS.	Month 13
4.3	Supplementing Deliverable 4.3 on the quality of AIS data with a comparison of the quality and coverage of the European AIS data from Dirkzwager and the Satellite data from LuxSpace. The satellite data will also be compared to the National AIS data from Greece.	Month 15
4.7	Report about possible new statistical output based on European AIS data. The report also describes analysed and elaborated scenarios for production of European and national statistics based on one single European data source of AIS data.	Month 16
4.8	Consolidated report on project results including a cost-benefit analysis of using AIS-data for official statistics.	Month 17

Milestones (SGA-2 only):

4.10	Progress and technical report of first internal WP-meeting	Month 9
4.11	Progress and technical report of the second internal WP-meeting	Month 14